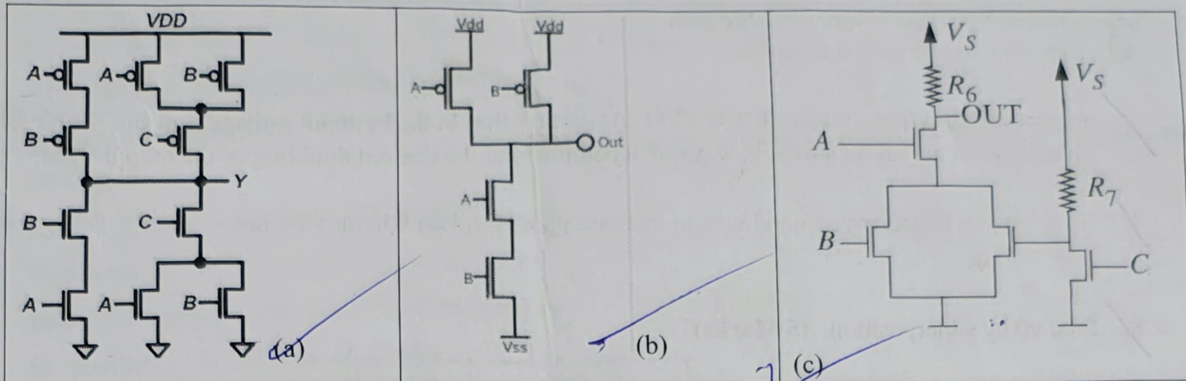
EE101 - Intro to Electrical and Electronic Circuits

MidSem (22/09/2025) – Course Instructor (Prof. Nagaveni)

Note: Assumptions should be written clearly to prevent any typo errors in the paper

Total Marks =50

1. Write the Boolean expression that describes the function of each of the circuits **(7 Marks)**



2. Write the CMOS logic for given Boolean expression $ab + c$ **(6 Marks)**
3. Consider a family of logic gates that operates under the static discipline with the following voltage thresholds: VOL = 0.5 V, VIL = 1.6 V, VOH = 4.4 V, and VIH = 3.2 V. **(3 Marks)**

logical 1 input

   a. What is the lowest voltage that can be output by an inverter for a logical 1 output?
   b. What is the highest voltage that must be interpreted by a receiver as a logical 0?
   c. What is the lowest voltage that must be interpreted by a receiver as a logical 1?

4. A driver has VOH = 3.5 V, VOL = 0.2 V. Driving a long cable causes VOH to drop by 0.8 V at the far end. Receiver thresholds: VIH = 2.6 V, VIL = 0.8 V. **(4 Marks)**
   a) Compute NM_H due to long cable cause (NM_H = Noise margin for High)
   b) Compute NM_L (NM_L=Noise margin for low)
   c) Is the high level still reliably received?

5. A system has two inputs A (student studied) and B (student attended class). The output should be **1 only if exactly one of them is true. (3 Marks)**
   a. Write the truth table
   b. Identify which basic gate directly implements it.

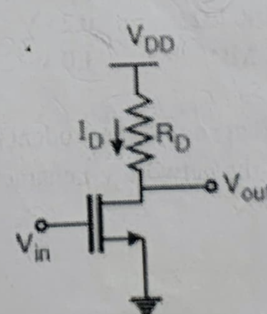
6. A lamp should glow if at least one of two switches (A or B) is ON **(3 Marks)**
   a. Identify gate.
   b. Write truth table.

7. Consider a **common-source MOSFET amplifier** shown below, with the following parameters: **(9 Marks)**

VDD=5V, RD=10kΩ,

MOSFET parameters:

k′(W/L)=1 mA/V^2, Vth=1V



DC/Large Signal Analysis

a. If the gate is biased such that VGS=2V, calculate the drain current ID.
b. Find the drain-to-source voltage VDS.
c. Verify whether the MOSFET operates in **saturation**. State the condition you checked.

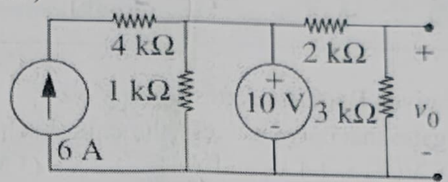
Small-Signal Analysis

d. Write the small signal model
e. Calculate the small-signal voltage gain.
f. Explain why the gain is negative.

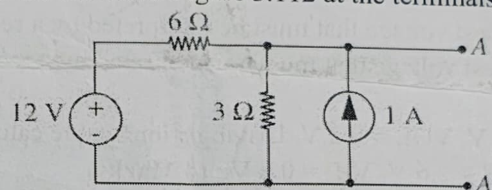
g. we desire an output voltage that is 2VO. Assuming that both the input voltage and the MOSFET do not change, what are all the possible ways of accomplishing the desired doubling of the output voltage?

h. Is increasing RD always a good way to increase gain? Explain why or why not.

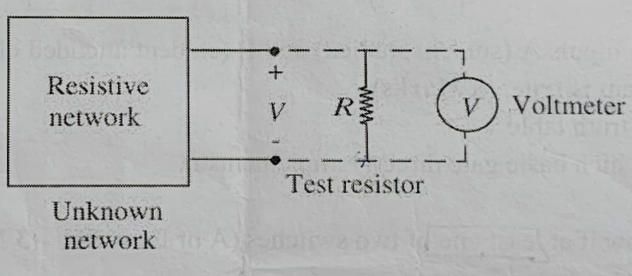
8. Find v0 by superposition. **(5 Marks)**



9. Find the Thévenin equivalent for the circuit in Figure 3.112 at the terminals AA' **(5 Marks)**



10. A student is given an unknown resistive network. She/He wishes to determine whether the network is linear, and if it is, what its Thévenin equivalent is. The only equipment available to the student is a voltmeter (assumed ideal), 100-k and 1-M test resistors that can be placed across the terminals during a measurement



The following data were recorded:

Test Resistor	Voltmeter Reading
Absent	1.5 V
100 kΩ	0.25 V
1 MΩ	1.0 V

What should the student conclude about the network from these results? Support your conclusion with plots of the network v-i characteristics. **(5 Marks)**

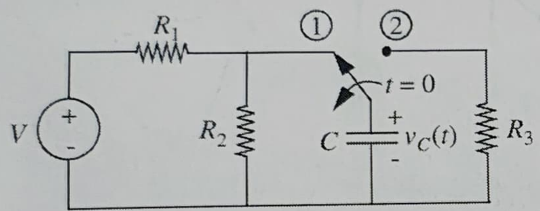
EE101 - Intro to Electrical and Electronic Circuits

Quiz-2 (31/10/2025) – Course Instructor (Prof. Nagaveni)

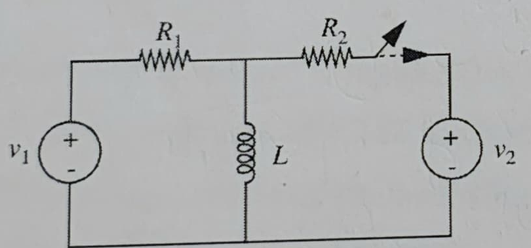
Note: Assumptions should be written clearly to prevent any typo errors in the paper

Total Marks =20 Marks

1. With the capacitor initially at rest (vC(0) = 0) and disconnected, the switch is closed to position (1) at time t = 0 in below Figure.
   i. Sketch the waveform vC(t) for t > 0. **[2Marks]**
   ii. Calculate the time constant. **[2Marks]**
   iii. At a time T > 0 (at least five time constants later), the switch is thrown (instantaneously) to position (2). Sketch vC(t) for t > T. **[2Marks]**
   iv. With R1 = R2 = R3, is the time constant in part (i) greater than, less than, or equal to the time constant in part (iii), justify? **[1 Mark]**

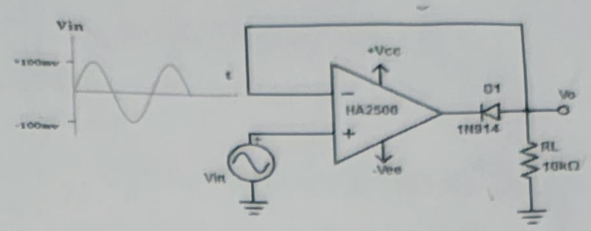


2. In the circuit shown in Figure, the switch opens at t = 0. Sketch and label iL(t) and vL(t). **(2+2 Marks)**

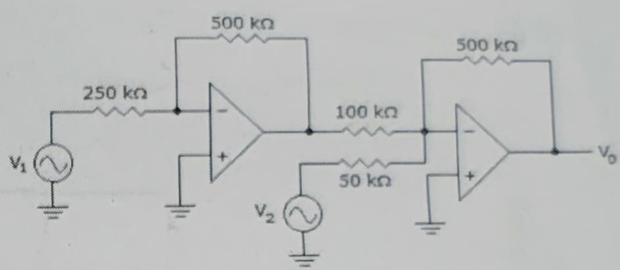


$v_1 = 5$ V $\quad v_2 = 3$ V, $\quad R_1 = 2$ kΩ, $\quad R_2 = 3$ kΩ,

3. Determine the output waveform of a negative small signal half wave rectifier. **(2Marks)**



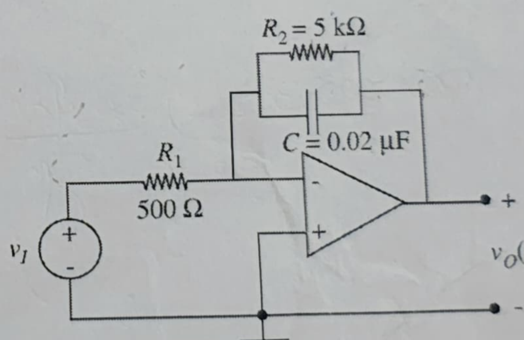
4. Calculate the output voltage if V1=V2 = 700 mV. **(3 Mark)**



5. An operational amplifier is connected as shown in Figure.

   a) What is the gain of the amplifier for ω = 0?. **[2 Marks]**

   b) Find the expression for Vo(jω)/Vi(jω) **[2 Marks]**



EE101 - Intro to Electrical and Electronic Circuits

EndSem (24/11/2025) – Course Instructor (Prof. Nagaveni)

Note: Assumptions should be written clearly to prevent any typo errors in the paper

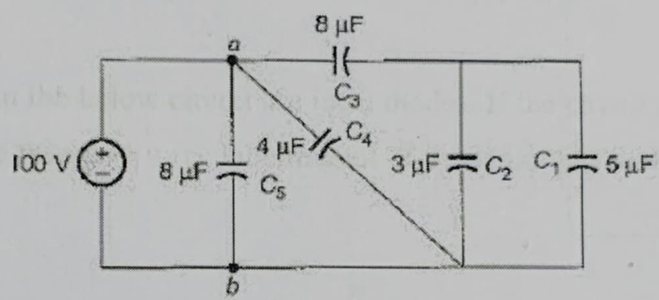
Total Marks = 70 Marks

1. A battery rated at 7.2 V and 10000 J is connected across a light bulb. Assume that the internal resistance of the battery is zero. Further assume that the resistance of the lightbulb is 100Ω.

   a. What is the power into the lightbulb? **(2 Marks)**

   b. How long will the battery last in the circuit? **(1 Mark)**

2. The charging time required to charge the equivalent capacitance between the given terminals a-b by a steady direct current of constant magnitude of 10 A is given by .... **(6 Marks)**



3. A 12-V, 115-Ah automobile storage battery is used to light a 6-W bulb. Assuming the battery to be a constant-voltage source, find how long the bulb can be lighted before the battery is completely discharged. **(2 Marks)**

4. A practical voltage source is represented by an ideal voltage source of 30 V along with a series internal source resistance of 1.2 Ω. Compute the smallest load resistance that can be connected to the practical source such that the load voltage would not drop by more than 2% with respect to the source open-circuit voltage. **(3 Marks)**

5. Let v(t) = Vmax cos ωt be applied to (a) a pure resistor, (b) a pure capacitor (with zero initial capacitor voltage, and (c) a pure inductor (with zero initial inductor current). Find the average power absorbed by each element. **(3 Marks)**

6. Consider the network shown in Figure

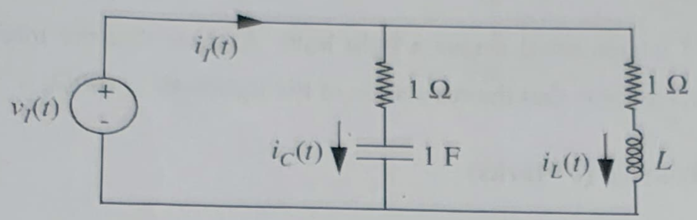
ith distance

1/2

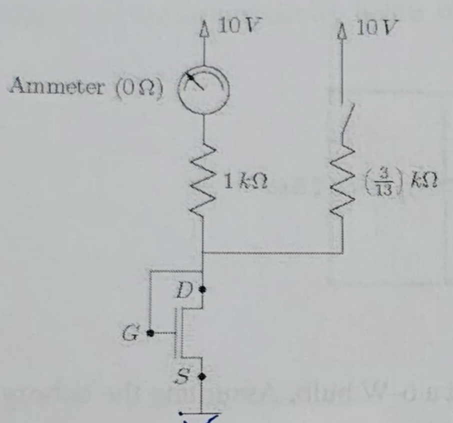
a) Show that by proper choice of the value of L, the impedance $V_i(s)/I_i(s) = Z_i(s)$ can be made independent of s. What value of L satisfies this condition? **(3 Marks)**

b) With L as determined in part (a), what is the value of Zi? **(1 Mark)**

c) Assume that the capacitor voltage and the inductor current are both zero for $t < 0$. Determine $i_C(t)$ equation when $V_I(t)$ is a unit step. **(3 Marks)**



7. In the below circuit, the current reading from the ammeter is 4 mA with the switch open and 3 mA with the switch closed, where $I_d = (K/2)(V_{gs}-V_T)^2$. What are V_T and K? **(5 Marks)**



8. Inlet Inc. manufactures two types of inverters – the SX model and the EX model – which satisfy the following static disciplines. **(6 Marks)**

Inverter SX satisfies a static discipline with the following voltage levels:

VOH = 4V VOL = 0.5V

VIH = 2V VIL = 1V

Similarly, Inverter EX satisfies a static discipline with these voltage levels:

VOH = 5V VOL = 0.7V

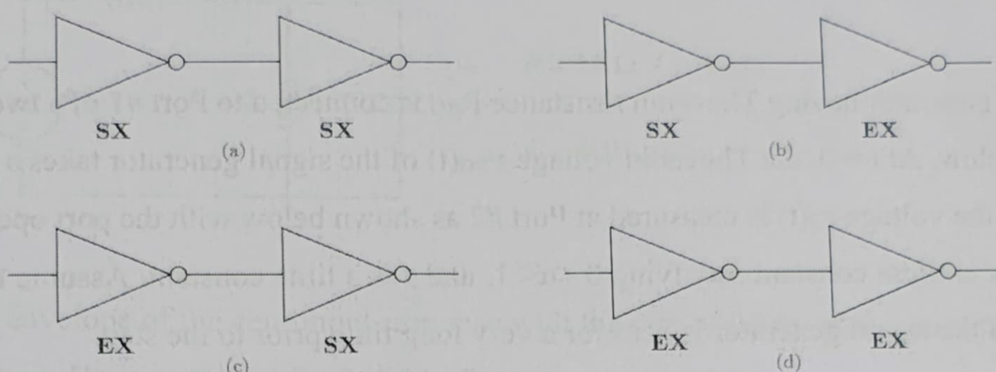
VIH =1.2V .VIL =1V

would like to bid on a contract for buffers issued by QuellCom. The buffers need to operate in a system which follows a static discipline with the following voltage levels:

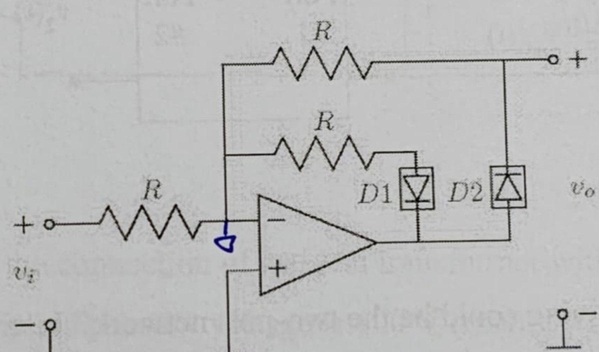
VOH = 4.55V VOL = 0.8V

VIH = 1.5V VIL = 1V

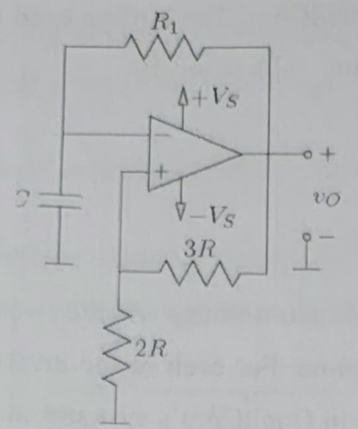
As an application engineer at Inlet you are tasked with determining whether a pair of inverters connected in series can satisfy QuellCom's static discipline. For each of the inverter pairs shown below, indicate whether the circuit can serve as a buffer in QuellCom's systems, and explain why.

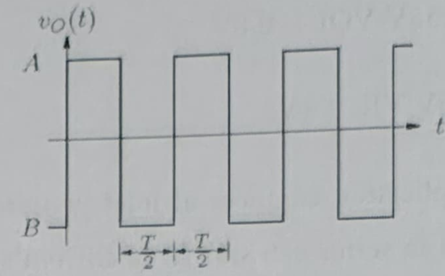


9. The diodes Dl and D2 in the below circuit are ideal diodes. If the circuit is driven by a signal $v_i(t)$ = *Vi cos(wt),* sketch the response $v_o(t)$, labelling all of the important features of the waveform. **(4 Marks)**

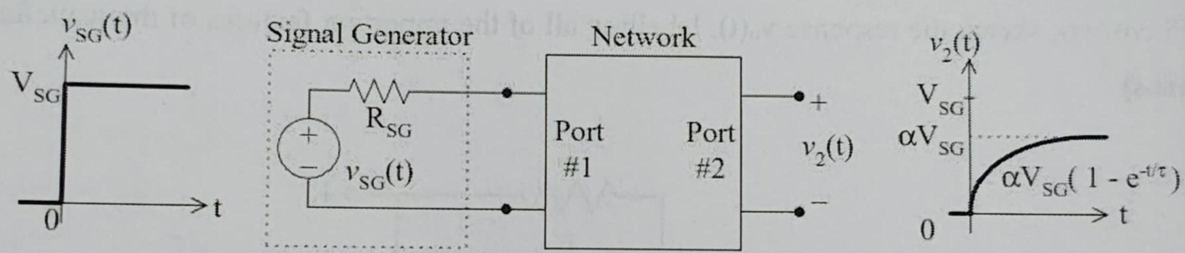


10. For the given circuit, the output wavform $vo(t)$ is illustrated in figure. The important features of the waveform are denoted A, B and T; please find the expression for these variables in terms of R1, R, C and power supply rails of the opamp which are +Vs and -Vs **(6 Marks)**

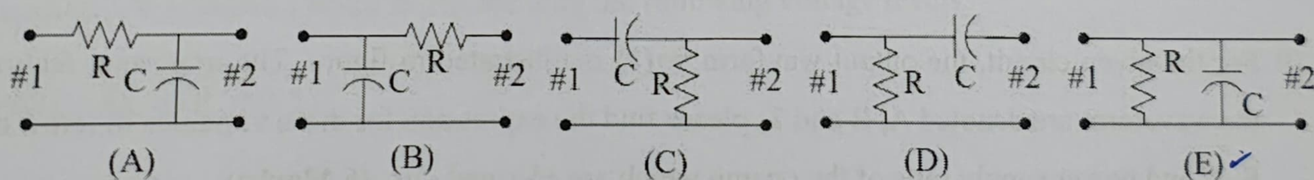




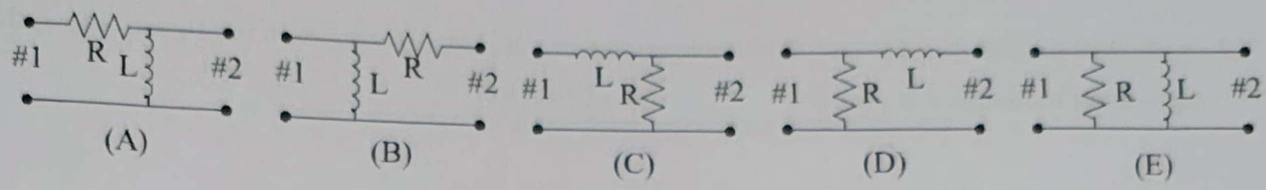
11. A signal generator having Thevenin resistance R_{SG} is connected to Port #1 of a two-port network as shown below. At $t = 0$, the Thevenin voltage $v_{SG}(t)$ of the signal generator takes a step from zero to V_{SG}, and the voltage $v_2(t)$ is measured at Port #2 as shown below with the port open-circuited. Note that α is a unitless constant satisfying $0 < \alpha < 1$, and τ is a time constant. Assume that the Thevenin voltage of the signal generator is zero for a very long time prior to the step.



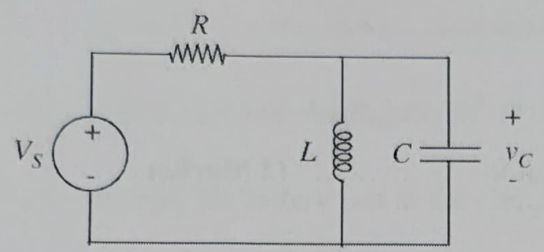
(11A) Which of the following could be the two-port network? Justify **(3 Marks)**



(11B) Which of the following could be the two-port network? Justify **(3 Marks)**



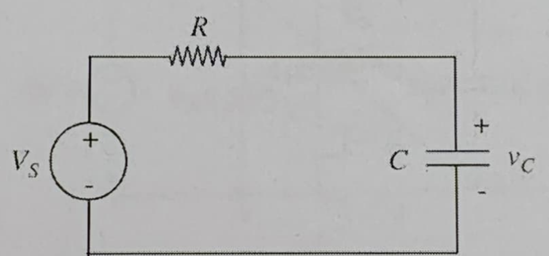
12. Is the zero input response of the circuit shown in Figure under-damped, over damped, or critically-damped? **(5 Marks)**



$R = 15\ \Omega$ $L = 1\ \mu H$

$C = 0.01\ \mu F$

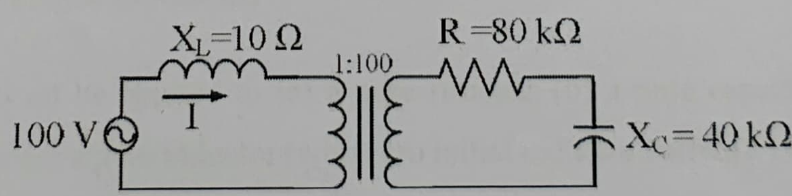
Compare the envelope of the zero input response with the rate of decay of the zero input response of the RC circuit. How do they differ? **(3 Marks)**



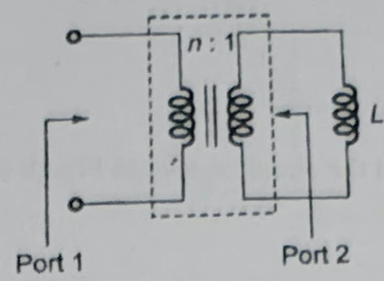
$R = 15\ \Omega$

$C = 0.01\ \mu F$

13. The following figure shows the connection of an ideal transformer with primary to secondary turns ratio of 1 : 100. The applied primary voltage is 100 V (rms), 50 Hz, AC. The rms value of the current I, in ampere, is __________. **(4 Marks)**



14. If an ideal transformer has an inductive load element at port 2 as shown in the figure below, the equivalent inductance at port 1 is …………… **(2 Marks)**



15. The Boolean expression $A \oplus B \oplus A$ is equivalent to …………… **(2 Marks)**

16. Output of the circuit shown below when S=1 and S=0 will be ……………… **(3 Marks)**